

Zhejiang Sci-Tech University

Undergraduate Thesis

**Accounting for Trade-Embodied Carbon in
China's Textile Industry and Analysis of Its
Determinants: An Input-Output Approach**

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Abstract

Against the backdrop of growing global attention to climate change and green trade, it has become increasingly important for China, the world's largest exporter of textiles, to delineate carbon-emission responsibilities in international trade on a scientifically defensible basis and quantify their determinants. Existing studies have explored accounting methods and determinants of embodied carbon in developing countries such as China, but important methodological and empirical gaps remain.

This study adopts an input-output approach to define embodied carbon in the textile industry and construct an accounting model. Using EXIOBASE 3, it calculates the carbon embodied in trade between China and five principal trading partners or regions from 2013 to 2022 and conducts a comparative analysis. The Logarithmic Mean Divisia Index (LMDI) is then used to examine the determinants of changes in net trade-embodied carbon, on the basis of which policy recommendations are proposed. The study develops the theoretical framework for industry-level embodied-carbon accounting and provides methodological and empirical support for a more equitable system of carbon-responsibility allocation.

The results show that: (1) the carbon-emission coefficient of China's textile industry declined from 71,343 to 18,747 kg CO₂/million EUR, but China remained a net exporter of embodied carbon; (2) the largest net balances in 2022 were with the United States and EU, at 0.8361 and 0.6681 Mt CO₂, while those with India and other Asian countries were 0.0583 and 0.1662 Mt CO₂; and (3) scale, efficiency, technology, and structural effects varied substantially over time. Reductions in trade with developed Western markets were associated principally with technological and structural improvement, whereas scale effects were more prominent in developing markets.

Keywords: green trade; carbon tariff; textile industry; trade-embodied carbon; input-output analysis; Logarithmic Mean Divisia Index

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Chapter 1

Introduction

1.1 Research Background and Significance

1.1.1 Research Background

As global warming becomes increasingly pronounced, carbon emissions and the associated greenhouse effect have emerged as major challenges shared by all countries. China has announced the goals of peaking carbon dioxide emissions before 2030 and achieving carbon neutrality before 2060 [1], bringing the methods used to account for carbon emissions and allocate responsibility into sharper focus. Against the rapid expansion of international trade, emissions generated during the production, processing, and transportation of goods are no longer confined within national borders but exhibit clear transnational and interregional characteristics. A substantial share of these emissions is released not in the country of final consumption but during production and export, prompting extensive discussion of embodied-carbon transfers through trade [2]. In supplying the global market, some developing countries bear environmental burdens and carbon liabilities commensurate with their export volumes. Given today's highly integrated global value chains, assigning carbon responsibility solely to producing countries is neither equitable nor scientifically robust, and challenges the conventional production-based framework for allocating emissions. The international community therefore needs a more rigorous and equitable accounting mechanism. Defining, accounting for, and identifying the determinants of embodied carbon is consequently important not only to theory, but also to global environmental governance and policymaking.

As the country with the world's largest concentration of manufacturing capacity, China has established an extensive export-oriented industrial system to serve international demand. This development has also imposed the dual pressures of resource consumption and environmental degradation. In carbon-intensive sectors in particular, large volumes of exported products are accompanied by substantial emissions, making China a central component of the global transfer of embodied carbon through trade. As green-trade principles expand and international mechanisms for allocating environmental responsibility mature, developed economies

such as the United States and the European Union have introduced policy instruments including carbon tariffs and carbon-footprint labels to constrain the carbon content of imports and promote global trade rules that account for environmental impacts. Export-oriented manufacturing economies such as China therefore face mounting pressure to reduce emissions, while the external impetus for green transformation continues to strengthen [3]. The textile industry is both a major pillar of China's traditional manufacturing sector and a comparatively energy- and carbon-intensive industry. Emissions occur throughout its value chain, from raw-material production and processing to manufacturing and export. As the industry becomes more deeply integrated into global supply chains, its importance in international production networks, and the associated cross-border carbon responsibility borne by China, continue to grow [4]. It is therefore increasingly important to define and quantify embodied carbon in China's textile industry in a manner that accurately reflects environmental impacts, and to decompose the factors driving its evolution.

Carbon emissions are currently quantified principally through life-cycle assessment (LCA) [5] and input-output analysis (IOA) [6]. LCA measures emissions over the life cycle of an individual product, from raw-material acquisition and production through use and final disposal, and is therefore suitable for micro-level environmental assessment. IOA, by contrast, examines material and energy flows among sectors within a national or industrial economic system and is more appropriate for macro-level emissions accounting. Although several studies have evaluated trade-related emissions between China and developed economies, systematic comparisons across multiple trading partners remain limited, and research focused specifically on the textile industry is insufficient. Both the breadth of coverage and the methodological framework therefore require further development. To address these limitations, this study uses an up-to-date database to construct a more comprehensive model of embodied carbon in the textile industry, compares multiple periods and regions, and applies index decomposition to identify the principal determinants of changes in trade-embodied carbon.

1.1.2 Research Significance

Using input-output analysis, this study systematically quantifies the embodied carbon generated by China's textile industry during production and related activities, and identifies the structural characteristics and cross-border transfer mechanisms of industrial emissions. Theoretically, the study extends the analytical framework for embodied emissions and trade-embodied carbon by adding an industry-specific dimension and a cross-country comparative perspective. It further develops the application of IOA to environmental accounting, particularly for modelling and pathway analysis in carbon-intensive, trade-surplus industries such as textiles. The introduction of the Logarithmic Mean Divisia Index (LMDI) method enables multidimensional identification of the relevant drivers, deepens understanding of the path-

ways and dynamics through which industrial emissions evolve, and provides a theoretical basis for partner-specific models.

From a practical perspective, amid the accelerating development of the global green-trade regime and the continued implementation of carbon border adjustment mechanisms, clarifying the direction of embodied-carbon flows can improve the transparency of China's trade-related emissions data and strengthen its voice in green-trade negotiations. Identifying the principal drivers by means of LMDI can also support more precisely targeted energy-conservation and emissions-reduction policies. The findings provide a quantitative basis for differentiated regional mitigation strategies and export-guidance policies, thereby supporting implementation of China's dual-carbon goals and green-industry strategy.

1.2 Literature Review on Trade-Embodied Carbon

1.2.1 Definitions of Embodied Carbon

The literature offers several formulations of embodied carbon. Liu et al. [6] define it as the sum of direct and indirect carbon dioxide emissions arising from energy consumption throughout the life cycle of a good or service, including production, processing, packaging, and transportation. Drawing on the United Nations Framework Convention on Climate Change, Lin et al. [4] define embodied carbon as all carbon emissions generated from raw-material acquisition, manufacturing, and transportation until the product is purchased by the final consumer. They also note alternative definitions that encompass the direct and indirect emissions generated during production and after-sales services, or the carbon dioxide emitted during the production, transportation, consumption, and disposal of any product. Mair et al. [3] similarly interpret embodied carbon as the emissions generated throughout production, transportation, consumption, and end-of-life treatment, emphasizing its value as an indicator of a product's actual environmental impact.

All of these definitions emphasize the environmental impacts generated over the complete life cycle. On this basis, the present study defines embodied carbon as the total direct and indirect carbon dioxide emissions generated throughout the life cycle of a product, including all emissions released during production, processing, packaging, transportation, consumption, and disposal.

1.2.2 Carbon-Accounting Methods for the Textile Industry

As a resource-intensive and export-oriented sector, the textile industry accounts for an important share of both production- and trade-related emissions worldwide. Scholars in China and elsewhere have extensively studied methods for accounting for its embodied carbon, producing a relatively systematic analytical framework. Current emissions accounting follows two

principal approaches: life-cycle assessment (LCA) [5] and input-output analysis (IOA) [6]. LCA is generally used to assess product-level carbon footprints, whereas IOA is better suited to macro-level studies of embodied carbon at the industry scale.

LCA systematically quantifies the emissions generated throughout the life cycle of an individual product, from raw-material acquisition and manufacturing to use and disposal, and can reveal how emissions are distributed among stages of textile production. Feng et al. [7], for example, examine the applicability of different LCA methods to textile carbon-footprint accounting. Zhao et al. [8] extend the analysis to virtual carbon and water flows in the global trade in denim products, thereby broadening the scope of LCA applications. However, LCA requires highly complete data and is generally product-specific. It is therefore difficult to extend across industries and regions, particularly when supply chains are complex, which limits both its applicability and scalability for embodied-carbon accounting.

IOA is a macro-level measurement tool capable of revealing quantitative relationships between economic activity and carbon emissions across countries, regions, and industries. It is particularly suitable for tracing the transfer of emissions through trade. Huang et al. [9], for example, use a single-region input-output model to calculate and assess trade-embodied carbon in 30 Chinese provinces. Qu et al. [10] introduce a minimum-entropy method to correct model errors, substantially improving the stability and reliability of the estimates. For the textile industry specifically, Liu et al. [6] construct an industry-appropriate accounting model and quantify the emissions induced in upstream and downstream industries by textile exports. They find that embodied emissions are concentrated primarily in electricity supply and raw-material processing. Lin et al. [4] further employ IOA to examine bilateral embodied carbon in China's foreign trade, emphasizing that China has long remained a net exporter of emissions through the global textile trade and consequently faces increasing pressure under international green-trade rules.

Despite the accumulated literature, three important limitations remain. First, most studies examine manufacturing as a whole rather than the textile industry in detail. Second, research on embodied carbon in textiles often focuses on a single year or a single bilateral relationship, such as China–US or China–EU trade. The absence of a systematic, multi-period and multi-region framework makes it difficult to reveal the dynamic evolution of industrial carbon-transfer patterns. Third, frequently used databases such as the World Input-Output Database (WIOD) are updated infrequently, while the input-output tables published by China's National Bureau of Statistics do not provide global coverage. Neither can always offer timely and accurate data amid the rapid restructuring of world trade and climate governance. A more frequently updated input-output database, such as EXIOBASE 3, is therefore needed to improve the accuracy and applicability of the results.

1.2.3 Research on Determinants

Changes in embodied carbon are affected by multiple factors. Existing assessments focus mainly on two complementary approaches. The first identifies the combined effects of scale expansion, structural adjustment, and technological progress at the macro level to reveal their systemic influence on emissions. The second uses mathematical models to decompose the relevant drivers quantitatively and determine the contribution and relative weight of each factor.

The literature generally regards the scale effect as the principal driver of growth in embodied carbon, particularly in export-led economies. Analysing China–US manufacturing trade, Cui [11] finds that although production efficiency improved, the large share of carbon-intensive industries prevented technological progress from fully offsetting the emissions growth caused by scale expansion. Han and Rong [12] further show that sustained export growth was the main driver of the increase in net trade-embodied carbon in China–India manufacturing trade. In addition to scale, industrial structure also has an important influence: a rising share of carbon-intensive sectors generally increases the embodied carbon per unit of output. Using an LMDI–PDA model, Hu and Wu [13] identify the contribution of technological change to declining carbon intensity in Chinese industrial sectors and demonstrate the positive role of technological progress in optimizing the energy mix and improving production processes. However, the technology effect is often constrained by an industry’s remaining mitigation potential and infrastructure, making it difficult to fundamentally alter short-term emissions trends. From the perspective of the digital economy, Yang et al. [14] find that although digital transformation can promote green technological progress, frequent equipment replacement and increased computational loads can also produce an energy rebound effect that indirectly raises emissions.

The Logarithmic Mean Divisia Index (LMDI) and Structural Decomposition Analysis (SDA) are the principal methods used to decompose and analyse these determinants. Owing to its perfect decomposition and absence of residual terms, LMDI is widely applied to dynamic analyses of emissions drivers. Numerous Chinese studies use the method to identify the marginal effects of export structure, energy efficiency, and technological substitution on changes in transferred carbon [11, 13]. SDA can characterize in detail the interactions among multiple factors under complex economic restructuring. Yang et al. [14], for example, employ SDA to quantify the association between development of the digital economy and changes in the embodied carbon of China’s foreign trade. International studies have also used SDA to extend comparative analysis of the interactions between developed and developing countries. Pu et al. [15] and Zhao et al. [16], respectively, reveal the profound effects of industrial structure and international price mechanisms on the embodied-carbon balance in the contexts of economic transition and China–US bilateral trade. SDA results, however, can be path-dependent and non-unique: different factor orderings may yield different results,

and the calculation is comparatively complex. By virtue of its path independence, reproducibility, and numerical stability, LMDI is more adaptable to multi-period, multi-country, and cross-industry datasets.

Substantial progress has therefore been made in identifying and quantifying the determinants of embodied carbon, and an initial analytical framework covering technological, structural, and scale variables has been established. Nevertheless, most studies still focus on manufacturing as a whole, leaving the textile industry insufficiently examined. Existing studies also tend to decompose gross embodied emissions rather than the net balance of trade-embodied carbon, and the factors selected for decomposition are not consistent across studies. Further research should analyse the determinants of changes in net trade-embodied carbon and introduce more representative factors so as to improve the relevance and feasibility of the resulting policy recommendations.

1.2.4 Critical Review of the Literature

Researchers broadly agree that embodied carbon captures environmental impacts across the complete life cycle. Its definition, however, has yet to be standardized, producing differences in accounting boundaries and statistical coverage among studies. Future work should therefore establish a consistent conceptual framework on the basis of clearly specified life-cycle boundaries and select accounting boundaries appropriate to the research objective, thereby improving comparability and applicability.

The preceding review also shows that research on embodied emissions in the textile industry has made progress in accounting methodology. LCA and IOA each offer distinct advantages: LCA is suitable for assessing the environmental impacts of individual products, whereas IOA is more appropriate for analysing embodied carbon across regions and industries. Several studies have constructed input-output-based accounting models for textiles and clarified the emissions responsibility borne by China as a manufacturing economy with a large trade surplus. Nonetheless, significant limitations remain. Research in China generally examines manufacturing as a whole or other energy-intensive sectors, with relatively little systematic work on textiles. Most findings concern bilateral relationships such as China–US or China–EU trade; a cross-country comparative framework has not yet been established, making it difficult to identify comprehensively the patterns and heterogeneity of embodied-carbon transfers between China and other economies. Moreover, many of the input-output databases used are outdated or narrowly focused and therefore cannot fully reflect recent changes in international trade. Future research should deepen accounting at the level of individual carbon-intensive industries such as textiles, employ updated data sources to capture rapidly changing trade patterns, and construct multinational networks of trade-embodied carbon to address the current lack of cross-country comparison.

Studies of determinants consistently identify the scale effect as the primary driver of increased embodied emissions. The capacity of technological progress and industrial restructuring to reduce emissions is constrained in countries with high concentrations of carbon-intensive industry. Although digitalization can facilitate the adoption of low-carbon technologies, it may in some circumstances also increase emissions. Existing work pays insufficient attention to the determinants of embodied carbon in bilateral trade, often considering only domestic industrial emissions or export-embodied emissions, and factor decompositions remain inconsistent. A multidimensional analysis is therefore required to capture interactions among industrial structure, energy efficiency, technological progress, digitalization, and other variables. Analysing the drivers of net trade-embodied carbon with a more standardized set of factors remains an important research direction. A comprehensive decomposition should account separately for imports and exports and for interactions among multiple drivers, thereby responding to the increasing complexity of international carbon governance.

1.3 Research Content and Methods

1.3.1 Research Content

This study conducts a quantitative analysis using IOA. It first defines embodied carbon and establishes its life-cycle boundary. Drawing on established accounting systems, it then develops a method suitable for the textile industry. The EXIOBASE database is used to calculate the embodied carbon of China's textile industry and the trade-embodied carbon between China and five major trading partners or regions. On this basis, the determinants of changes in net trade-embodied carbon are analysed and corresponding policy recommendations are proposed.

The research comprises three principal components:

- (1) **Definition and accounting methodology.** On the basis of a consistent life-cycle boundary and emissions scope, the object of embodied-carbon accounting is clearly specified and distinguished from conventional carbon emissions. An IOA-based accounting model suitable for cross-country comparison is then constructed.
- (2) **Accounting and analysis.** Carbon-emission coefficients and total embodied emissions are calculated with the input-output model and combined with import and export data to quantify trade-embodied carbon between China's textile industry and its principal trading partners. Cross-country comparisons are used to examine the emissions responsibility borne by China in global supply chains and the transfer of emissions from producing to consuming countries through international trade.
- (3) **Selection and index analysis of determinants.** Seven determinants of changes in net trade-embodied carbon are examined: the energy-mix effect, energy-efficiency ef-

fect, production-technology effect, export-structure effect, export-scale effect, import-structure effect, and import-scale effect. Their diagnostic contributions are estimated using an LMDI-based index framework and inform the policy discussion.

1.3.2 Research Methods

The following methods are employed:

- (1) **Environmentally extended input-output analysis.** EXIOBASE's multi-regional input-output tables are combined with environmental extensions and final-demand data to calculate production emissions and coefficient-based estimates of the carbon embodied in textile trade.
- (2) **Comparative analysis.** Building on the accounting results, the carbon intensity embodied in exports and the transfer of emissions between China and its principal trading partners are compared to identify differences in carbon responsibility.
- (3) **Logarithmic Mean Divisia Index (LMDI).** Changes in trade-embodied carbon between consecutive years are assessed using seven index effects: energy mix, energy efficiency, production technology, export structure, export scale, import structure, and import scale.

1.3.3 Technical Route

The technical route adopted in this study is shown in Figure 1.1.

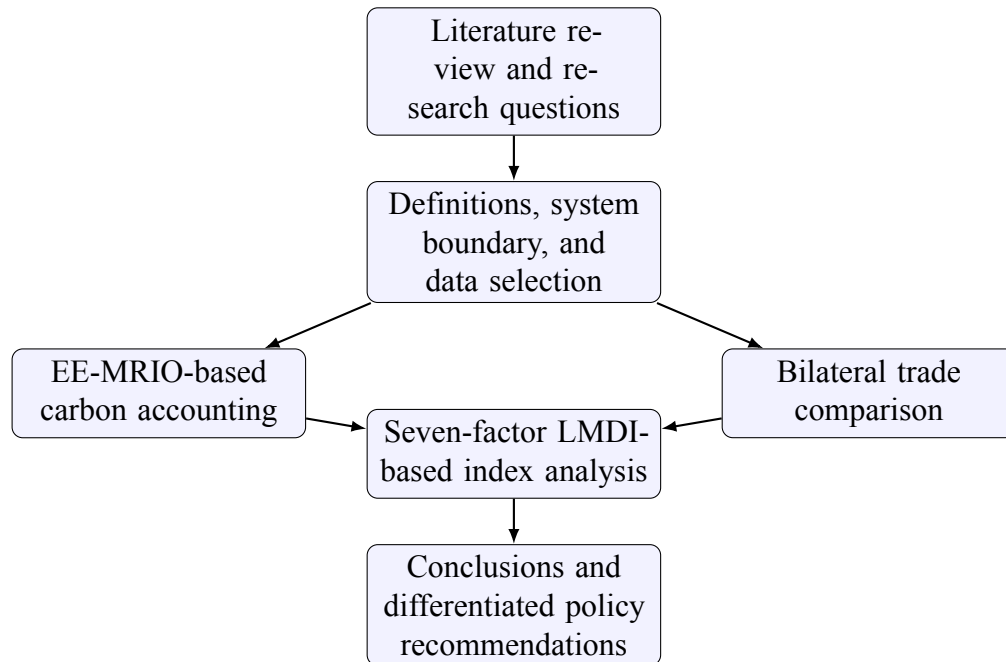


Figure 1.1: Technical route of the study

1.4 Contributions

The principal contributions of this study are as follows:

- (1) The accounting object is narrowed to the textile industry, departing from the conventional use of national economies or manufacturing as a whole as the unit of analysis. An input-output accounting model is developed for a specific industry. The analysis of trade-embodied carbon adopts a multi-country perspective rather than the China–US or China–EU bilateral paradigms common in earlier work. Five major partner groups—the United States, Japan, India, the European Union, and other Asian countries—are selected to construct a cross-regional and multidimensional analytical framework.
- (2) An LMDI-based index framework is used to classify and estimate seven determinants associated with changes in net trade-embodied carbon. By accounting for both sides of bilateral trade, the resulting model incorporates energy mix, energy efficiency, production technology, import and export structure, and import and export scale.

Chapter 2

Input-Output Accounting for Trade-Embodied Carbon in the Textile Industry

The preceding chapter reviewed the theoretical foundations and practical applications of embodied-carbon research in the textile industry, established IOA as the core methodological approach, and identified gaps in the research object, data foundation, and analytical dimensions of previous studies. This chapter presents the empirical accounting framework for trade-embodied carbon in China's textile industry, focusing on definitions, accounting methods, and model construction.

2.1 Definitions of Embodied Carbon and Trade-Embodied Carbon in the Textile Industry

As discussed in Chapter 1, definitions of embodied carbon differ across studies, but all emphasize environmental impacts over the complete life cycle. This study defines embodied carbon as the total direct and indirect carbon dioxide emissions generated throughout the life cycle of a product, including emissions released during production, processing, packaging, transportation, consumption, and disposal. Embodied carbon in the textile industry thus refers to the carbon dioxide emitted throughout the life cycle of textile products, from production and processing to end-of-life treatment.

Trade-embodied carbon is the embodied carbon associated with traded goods and encompasses export-embodied emissions, import-embodied emissions, and the net balance of trade-embodied carbon [14]. Export-embodied emissions are generated in producing goods for foreign markets, whereas import-embodied emissions are associated with goods imported for domestic use. Net trade-embodied carbon is the difference between the two and represents the embodied-carbon relationship between a pair of trading economies.

2.2 Input-Output Model

2.2.1 Environmentally Extended Input-Output Framework

The analysis draws on the environmentally extended multi-regional input-output (EE-MRIO) structure of EXIOBASE. By describing technical relationships among industries and the composition of final demand, the framework establishes a quantitative relationship between economic activity and carbon emissions. Country- and sector-specific blocks are then extracted for the empirical calculations. Table 2.1 presents the basic input-output relationships in a hypothetical economy with n industrial sectors. The total output of a sector consists of intermediate demand from producing sectors and final demand from end users.

Table 2.1: Basic structure of an input-output table

Inputs \ Outputs		Intermediate products				Final demand	Total output
		Sector 1	Sector 2	...	Sector n	Y_i	X_i
Intermediate inputs	Sector 1	x_{11}	x_{12}	...	x_{1n}	y_1	X_1
	Sector 2	x_{21}	x_{22}	...	x_{2n}	y_2	X_2
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$
	Sector n	x_{n1}	x_{n2}	...	x_{nn}	y_n	X_n
Value added		v_1	v_2	...	v_n		
Total inputs		X_1	X_2	...	X_n		

2.2.2 Accounting Model for Trade-Embodied Carbon

2.2.2.1 Input-output model and embodied-carbon accounting

The calculation proceeds in five steps.

1. Carbon-emission coefficient. For sector j , the carbon-emission coefficient is obtained from its carbon emissions C_j and output X_j :

$$e_j = \frac{C_j}{X_j}, \quad (2.1)$$

where e_j is measured in kilograms of CO₂ per million euros of output.

2. Technical-coefficient matrix. The matrix $\mathbf{A} = [a_{ij}]$ describes input-output relationships among industries, where

$$a_{ij} = \frac{x_{ij}}{X_j}. \quad (2.2)$$

Each element a_{ij} represents the input from sector i required per unit of output in sector j , thereby capturing interindustry dependence.

3. Output and final-demand vectors. Let $\mathbf{X} = (X_1, \dots, X_n)^T$ denote total sectoral output, measured in million euros, and let $\mathbf{Y}_{\text{final}}$ denote final demand, including consumption, investment, and exports. Export and consumption demand are of particular relevance to embodied-carbon accounting.

4. Leontief inverse. The Leontief inverse $(\mathbf{I} - \mathbf{A})^{-1}$ captures the direct and indirect production required across sectors to satisfy one unit of final demand. Here $\mathbf{I}$ is the identity matrix.

5. Total embodied emissions. Total embodied emissions are calculated by multiplying the direct carbon-emission coefficient by the total production requirements determined by the Leontief inverse and by final demand:

$$F = \mathbf{e}^T (\mathbf{I} - \mathbf{A})^{-1} \mathbf{y}_{\text{final}}, \quad (2.3)$$

where F denotes total embodied emissions, measured in kilograms of CO_2 ; $\mathbf{e}$ is the vector of direct sectoral carbon-emission coefficients; and $\mathbf{y}_{\text{final}}$ is the final-demand vector, including consumption, investment, and exports. This vector form corrects the dimensional relationship among the emissions coefficients, Leontief inverse, and final demand.

2.2.2.2 Accounting for trade-embodied carbon

1. Export-embodied emissions. For the bilateral estimates reported in this study, economy- and sector-specific carbon coefficients are applied to the corresponding final-demand trade values. Export-embodied emissions are calculated as the product of the exporting sector's coefficient and export value:

$$F_{\text{export}} = e_j X_{\text{export}}, \quad (2.4)$$

where F_{export} is the embodied carbon in China's exports from sector j , measured in kilograms of CO_2 , and X_{export} is the value of those exports.

2. Import-embodied emissions. The coefficient used for imports is calculated for the corresponding sector in the source economy s :

$$e_s = \frac{C_s}{X_s}. \quad (2.5)$$

Import-embodied emissions are then:

$$F_{\text{import}} = e_s X_{\text{import}}, \quad (2.6)$$

where F_{import} is the embodied carbon in China's imports from the corresponding foreign sector, e_s is that sector's carbon-emission coefficient, and X_{import} is the value of China's imports from the source economy.

3. Net trade-embodied carbon. The net balance is the difference between export- and import-embodied emissions:

$$F_{\text{net}} = F_{\text{export}} - F_{\text{import}}. \quad (2.7)$$

The variable F_{net} represents the net trade-embodied carbon of China's textile industry and indicates the industry's carbon transfer through global trade. A positive value denotes a net export of embodied carbon; a negative value denotes a net import.

2.3 Data Sources and Processing

2.3.1 Data Sources

Databases commonly used for international environmental-economic analysis include EX-IOBASE, the World Input-Output Database (WIOD), EORA, the Global Trade Analysis Project (GTAP) database, and national input-output tables such as those published by China's National Bureau of Statistics. These sources differ in geographical coverage, sectoral resolution, and environmental extensions and are widely used to study global value chains, trade structures, and embodied carbon. Some, however, suffer from outdated observations, limited coverage, or missing data. WIOD, for example, extends only to 2016, while China's national input-output tables omit several years and cover only the domestic economy.

Following Yang et al. [14], this study therefore uses EXIOBASE 3 [18] as its principal data source. EXIOBASE is an environmentally extended multi-regional input-output database developed jointly by institutions including the Netherlands Environmental Assessment Agency (PBL), Vienna University of Economics and Business (WU), and the Wuppertal Institute in Germany. It offers broad geographical coverage, detailed classifications, and a complete time series. The database provides input-output data for 44 countries and five rest-of-world regions, covering 163 industries and 200 product groups, together with detailed satellite accounts for carbon emissions, energy consumption, water use, and other environmental pressures. It is consequently well suited to embodied-carbon accounting and analysis of international green trade.

The version used is EXIOBASE 3.9.5 [18], released on 14 February 2025. At the time of the original study, this was the latest publicly available release and provided strong temporal

continuity and internal consistency. Relative to other databases, EXIOBASE offers a more detailed classification of textile-related activities and broader coverage of relevant emissions. The analysis covers the ten years from 2013 to 2022. From 2013 onward, EXIOBASE exhibits comparatively stable industrial classifications, environmental-account structures, and regional consistency, supporting reliable intertemporal comparison. This interval also encompasses important changes in the export structure of China’s textile industry, the implementation of emissions policies, and the configuration of international trade. It can therefore represent the dynamic evolution of trade-embodied carbon with reasonable completeness. The year 2022 is the latest available observation in this database version, ensuring the timeliness of the dataset used in the study.

2.3.2 Data Processing

The principal export markets for Chinese textiles include ASEAN, the United States, the European Union, and Japan [21]. The empirical analysis selects the United States, Japan, India, the European Union, and other Asian countries for cross-sectional comparison. EU member states are aggregated into a single region during data processing, as shown in Table 2.2.

Table 2.2: Geographical scope of the study

Economy or region	Code	Countries or regions included (database codes)
China	CN	CN
United States	US	US
Japan	JP	JP
India	IN	IN
Other Asian countries	WF	WF
European Union	EU	AT, BE, BG, HR, CY, CZ, DK, EE, FI, FR, DE, GR, HU, IE, IT, LV, LT, LU, MT, NL, PL, PT, RO, SK, SI, ES, SE

The textile-related industries listed in Table 2.3 cover the principal stages of the textile value chain, from raw-material acquisition and primary processing to finished-product manufacturing and waste treatment. They represent the industry’s dependence on environmental resources across the complete life cycle and establish the system boundary for identifying and accounting for emissions pathways. These industries are aggregated as the textile industry for the purposes of this study.

In accordance with the definition of embodied carbon, all entries in the EXIOBASE emissions inventory containing CO₂ were screened. They are listed in Table 2.4.

To ensure an industry-specific system boundary and select appropriate emissions pathways, the study follows the theoretical framework developed by Lenzen et al. [22] for constructing CO₂ multipliers and classifying emissions pathways in multi-regional input-output models. Their framework requires carbon sources to be selected according to the actual emissions char-

Table 2.3: Textile-related industries included in the study

Industry	EXIOBASE sector
Cultivation of plant-based fibres	Cultivation of plant-based fibers
Wool and silkworm-cocoon production	Wool, silk-worm cocoons
Textile manufacturing	Manufacture of textiles (17)
Manufacture of apparel; dressing and dyeing of fur	Manufacture of wearing apparel; dressing and dyeing of fur (18)
Tanning and dressing of leather; manufacture of luggage, handbags, saddlery, harness, and footwear	Tanning and dressing of leather; manufacture of luggage, handbags, saddlery, harness and footwear (19)
Incineration of textile waste	Incineration of waste: Textiles
Landfilling of textile waste	Landfill of waste: Textiles

Table 2.4: CO₂ emission entries in EXIOBASE

Emission category	Database code
Carbon dioxide from combustion, emitted to air	CO2 - combustion - air
Biogenic carbon dioxide from combustion, emitted to air	CO2_bio - combustion - air
Carbon dioxide from non-combustion processes in cement production, emitted to air	CO2 - non combustion - Cement production - air
Carbon dioxide from non-combustion processes in lime production, emitted to air	CO2 - non combustion - Lime production - air
Carbon dioxide from peat decay in agriculture, emitted to air	CO2 - agriculture - peat decay - air
Fossil carbon dioxide from waste treatment, emitted to air	CO2 - waste - fossil - air
Biogenic carbon dioxide from waste treatment, emitted to air	CO2 - waste - biogenic - air

acteristics of the production activities under investigation and excludes pathways unrelated to the target industry, such as peat oxidation in agriculture and non-combustion processes in building-material production. This approach constrains the system boundary and improves interpretability. Following a comprehensive review of the EXIOBASE environmental accounts and recognizing that textile emissions are concentrated in energy combustion, some forms of biomass combustion, and waste treatment [23], four direct-emission categories are retained:

- (1) **Energy-combustion emissions (C02 – combustion – air).** According to the EXIOBASE definition, this category includes carbon dioxide released directly by burning fossil fuels such as coal, petroleum, and natural gas in production, as well as upstream combustion used to supply heat, power, or electricity. Principal sources include fossil-fuel combustion in industrial boilers, power stations, motor vehicles, and gas-fired furnaces. It is a core component of embodied-carbon accounting and is included in full.
- (2) **Biogenic combustion emissions (C02_bio – combustion – air).** These are carbon dioxide emissions released by the combustion of renewable biomass energy sources such as wood, straw, vegetable oil, and biogas. Because the study adopts a life-cycle perspective, this category is selectively included.
- (3) **Fossil waste-treatment emissions (C02 – waste – fossil – air).** This category covers carbon dioxide released when solid waste containing fossil carbon—including plastics, rubber, and synthetic fibres—is treated, for example by incineration. The emissions arise principally from the incineration of industrial waste, plastic garments, chemical-fibre products, and thermal recovery from industrial waste. They must be included, particularly for the end-of-life treatment of textiles containing polyester and other polymer fibres.
- (4) **Biogenic waste-treatment emissions (C02 – waste – biogenic – air).** These are emissions released during the incineration or treatment of biogenic waste such as cotton, bast fibres, and wood-based materials. Principal sources include the incineration of cotton and linen garments, wood-processing residues, and biodegradable natural-fibre waste. This category is also selectively included.

After selecting the geographical scope, textile-related industries, and CO₂ entries, the data are harmonized across time, industrial classifications, and units to ensure accurate and comparable results.

- (1) For the temporal dimension, the completeness and consistency of all indicators are checked for each year from 2013 to 2022. This ensures that the input-output tables

and environmental satellite accounts remain continuous and structurally consistent, as required for time-series and index-decomposition analysis.

- (2) Following the sector classification in EXIOBASE 3, the seven textile-related industries are aggregated into a single analytical unit. This improves the concentration and representativeness of the sectoral estimates and avoids fragmented results.
- (3) All selected emissions indicators are originally expressed in kilograms. Million tonnes (Mt) are used where necessary to avoid unwieldy values, and the applicable unit is stated for each result. The selected indicators cover combustion, waste treatment, and other relevant sources. Other greenhouse gases, including CH₄, N₂O, and HFCs, are excluded, and the data are not converted to carbon-dioxide equivalents. The analysis consequently focuses on cross-border transfers and responsibility for carbon dioxide itself.
- (4) EXIOBASE monetary values are denominated in million euros. To avoid errors arising from exchange-rate fluctuations, the original currency unit is retained when calculating carbon intensity, such as emissions per unit of output. This preserves internal consistency and the relative stability of the results.

2.4 Chapter Summary

This chapter established the accounting methodology for trade-embodied carbon in the textile industry. It defined embodied carbon and trade-embodied carbon and constructed an accounting framework using environmentally extended input-output data. The empirical dataset comprises ten years of EXIOBASE 3 observations from 2013 to 2022. Seven textile-related industries are aggregated, the relevant carbon pathways are identified, and the principal emissions entries are selected to establish a clearly specified system boundary. The resulting model and data-processing procedure provide the methodological and empirical foundation for the analysis in the following chapters.

Chapter 3

Accounting Results and Analysis

Having established the accounting method and data foundation, this chapter presents the empirical estimates for China's textile industry from 2013 to 2022 and analyses their structural characteristics and underlying causes.

3.1 Import and Export Trade in the Textile Industry

The analysis begins with trends in the industry's aggregate imports and exports from 2013 to 2022 and then examines the bilateral trade pattern with the five focal economies or regions.

Table 3.1: Imports and exports of China's textile industry (million EUR)

Year	Total exports	Export share	Total imports	Import share	Trade balance	Five leading export destinations
2013	142137.1806	0.1491	68526.1762	0.0719	73611.0043	US, EU, RU, GB, WF
2014	153289.3941	0.1618	75521.4318	0.0797	77767.9623	US, EU, GB, RU, WF
2015	184887.1672	0.1443	76256.3586	0.0595	108630.8086	US, EU, GB, WF, RU
2016	161870.4785	0.1191	89557.9772	0.0659	72312.5013	US, EU, GB, WF, RU
2017	180058.9921	0.1270	50654.2699	0.0357	129404.7222	US, EU, GB, WF, JP
2018	177113.8810	0.1242	76355.1072	0.0535	100758.7738	US, EU, JP, WF, GB
2019	190750.0747	0.1133	102937.0311	0.0611	87813.0436	US, EU, WF, GB, JP
2020	179468.6343	0.1207	108205.6412	0.0728	71262.9931	US, EU, GB, WF, CA
2021	203122.2385	0.0998	148028.8147	0.0727	55093.4238	US, EU, JP, GB, WF
2022	188062.3433	0.0912	146339.6094	0.0710	41722.7339	US, EU, GB, WF, JP

China's textile industry maintained a trade surplus throughout the ten-year period. As shown in Table 3.1, exports increased from EUR 142,137.18 million in 2013 to EUR 188,062.34 million in 2022, equivalent to a compound annual growth rate of approximately 3.16%. Despite the increase in value, the share of exports in total industry output declined overall from 14.91% to 9.12%, although it rose in several individual years. This pattern indicates that the industry's dependence on foreign markets weakened in relative terms over the full study period.

Imports increased from EUR 68,526.18 million in 2013 to EUR 146,339.61 million in 2022, a considerably faster rate than exports. Their share of industry output fluctuated, reaching a low in 2017 before recovering. Although the industry remained in surplus, the balance narrowed from EUR 73,611.00 million in 2013 to EUR 41,722.73 million in 2022. This contraction suggests that the net export of embodied carbon may also have declined over the long term as the export share fell.

The destination structure of textile exports remained concentrated. The United States, European Union, United Kingdom, and other Asian countries appeared frequently among the leading destinations, while Japan, Russia, and Canada entered the top five in particular years. Although broadly stable, the export portfolio also exhibited a modest trend towards diversification.

Appendix A shows that the United States remained China's largest textile export market. Exports to the United States amounted to EUR 35,698.62 million in 2013, producing a bilateral surplus of EUR 31,956.72 million. By 2015 exports had risen to EUR 49,222.91 million and the surplus to EUR 45,792.09 million, demonstrating the highly asymmetric division of production between the two economies. Trade with the European Union also generated a stable surplus: in 2015, China exported EUR 29,124.24 million of textile products to the EU and imported EUR 5,991.92 million, for a surplus of EUR 23,132.33 million.

Trade with Japan and India was more complex. Japan remained an important export market, but its exports to China were also substantial and China recorded a deficit in some recent years. China's exports to India were comparatively small between 2013 and 2015, while imports fluctuated considerably, resulting in a limited overall surplus. Exports to other Asian countries (WF) increased from EUR 4,184.72 million in 2013 to EUR 7,220.70 million in 2015, accompanied by a marked increase in the surplus. This pattern indicates the growing importance of other Asian markets in China's textile trade during the period.

Over the decade, therefore, China's textile trade was export-led, consistently in surplus, and concentrated but gradually diversifying. This pattern directly underpins China's position as a net exporter of embodied carbon. Changes in the composition of imports and exports also provide the empirical basis for the subsequent accounting and decomposition analyses.

3.2 Embodied Carbon in the Textile Industry

Substituting the relevant EXIOBASE observations into the model yields annual carbon-emission coefficients for China's textile industry. These coefficients directly measure changes in embodied-carbon intensity and provide the technical basis for the ensuing analysis.

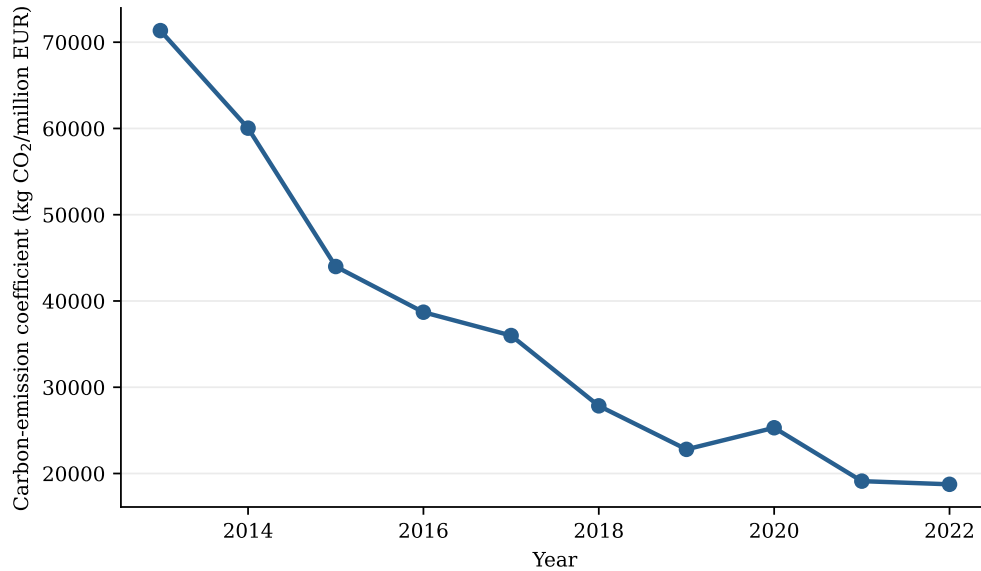


Figure 3.1: Carbon-emission coefficient of China's textile industry

As Figure 3.1 shows, the carbon-emission coefficient declined markedly between 2013 and 2022. It was approximately 71,343 kg CO₂/million EUR in 2013. By 2022, it had fallen to 18,747 kg CO₂/million EUR, a cumulative reduction of more than 70%. This indicates substantial progress in technology renewal, energy efficiency, and emissions control.

The coefficient fell particularly rapidly between 2013 and 2018, coinciding with accelerated development of China's green-manufacturing system and adjustment of its energy mix. In 2015, the State Council issued *Made in China 2025*, which explicitly called for the green transformation and upgrading of manufacturing. The Ministry of Industry and Information Technology subsequently released the *Industrial Green Development Plan (2016–2020)*, setting a target of reducing energy consumption per unit of industrial value added by 18%, promoting green processes, and eliminating obsolete capacity. The National Development and Reform Commission also issued implementation guidance on electricity-market development, promoting market-based trading of clean energy.

After 2019, the rate of decline slowed, and the coefficient increased slightly in 2020. This reversal coincided with the COVID-19 pandemic [24], when disruptions to industrial supply chains, interrupted operations, and restart costs affected production efficiency. The available data do not isolate the causal contribution of each short-term mechanism, so the 2020 movement should be interpreted cautiously.

Overall, however, the industry continued to shift towards lower-carbon and cleaner production, providing a favourable basis for reducing the emissions intensity of embodied carbon. The trend also indirectly indicates the potential role of technological change in driving embodied-carbon dynamics. As the carbon burden per unit of output declines, emissions growth can be partly mitigated even if aggregate output continues to expand.

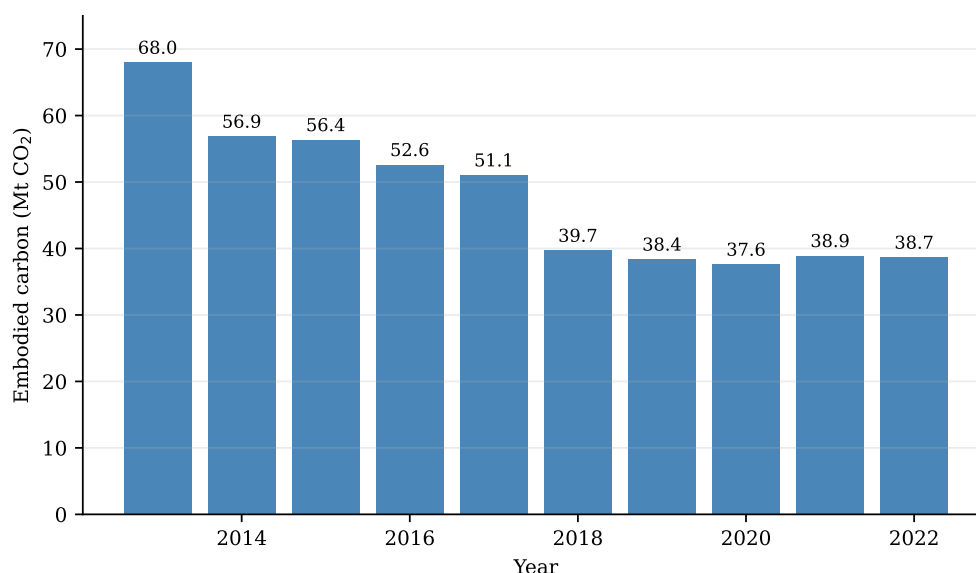


Figure 3.2: Total embodied carbon of China's textile industry

Whether total emissions improved in parallel with the declining coefficient is an important measure of the industry's low-carbon transition. Figure 3.2 shows that total embodied emissions fell gradually from a high level and then stabilized between 2013 and 2022.

Total embodied emissions were approximately 68 Mt CO₂ in 2013 and declined to about 39.7 Mt CO₂ in 2018, a cumulative reduction of approximately 41.6% over five years. From 2019 onward, the total stabilized and fluctuated modestly around 38 Mt CO₂. Although emissions intensity continued to decline, the rate of improvement diminished. Basic energy-efficiency retrofits may therefore be approaching saturation, while the potential for structural mitigation has yet to be fully realized.

3.3 Trade-Embodied Carbon

3.3.1 Comparison of Carbon-Emission Coefficients

Table 3.2 reports the calculated carbon-emission coefficients of the six economies or regions from 2013 to 2022.

Table 3.2: Carbon-emission coefficients by economy or region (kg CO₂/million EUR)

Year	CN	EU	IN	JP	US	WF
2013	71343.0111	30823.0070	72127.9529	65055.6625	88758.6081	31992.0210
2014	60035.9498	28118.1852	107942.9686	76978.4343	77591.8449	31000.0723
2015	43994.1773	26727.9569	52716.6897	45365.0520	58831.8934	48240.9126
2016	38699.0925	27618.8890	59166.0900	40565.4913	54903.5073	115166.7876
2017	35994.7112	28437.1564	60829.4021	34439.9204	53733.8790	64048.6422
2018	27842.9277	25487.4272	52871.1894	44047.0583	54313.3525	36799.4796
2019	22798.4906	26855.7604	43640.7146	40217.3070	50859.3878	26573.7399
2020	25302.8745	30650.6325	42689.7742	212796.3685	45764.8303	28322.2472
2021	19114.4811	17510.6693	31650.0991	57634.1469	22040.5043	31258.9112
2022	18746.9172	16179.9056	31092.3535	59026.6288	22788.6179	32649.9234

The coefficients generally declined over the decade, reflecting a common global trend towards lower-carbon production and reduced energy use. Their levels and trajectories, however, differed substantially.

China's coefficient stood at 71,343.01 kg CO₂/million EUR in 2013, a moderately high level. As the domestic green-manufacturing system developed, it declined sharply after 2018 and reached 18,746.92 kg CO₂/million EUR in 2022, a reduction of more than 73% and evidence of sustained improvement in carbon intensity.

India exhibited one of the highest coefficients. It increased from 72,127.95 in 2013 to 107,942.97 in 2014 (both in kg CO₂/million EUR), between 1.5 and 4 times the contemporaneous levels in several other regions. Although it subsequently declined, it remained at 31,092.35 kg CO₂/million EUR in 2022. This comparatively high level is consistent with the large share of fossil fuels in India's energy mix and its gradual transition to cleaner production [12].

The European Union maintained comparatively low coefficients throughout the period. Its value fell from 30,823.01 in 2013 to 16,179.91 kg CO₂/million EUR in 2022, making it one of the regions with the strongest mitigation performance.

Japan's coefficient was 65,055.66 kg CO₂/million EUR in 2013, slightly lower than those of China, the United States, and India. It declined markedly from 2013 to 2017, but subsequently increased and reached 59,026.63 kg CO₂/million EUR in 2022. The exceptional spike to 212,796.37 kg CO₂/million EUR in 2020 is an outlier and should be interpreted cautiously because temporary output changes or aggregation effects can amplify coefficient estimates. The trajectory therefore shows greater volatility and a smaller overall reduction than China's. The United States displayed a fluctuating decline from a high initial level. Its coefficient was 88,758.61 kg CO₂/million EUR in 2013, the highest among the regions that year, and fell to 22,788.62 kg CO₂/million EUR in 2022, a reduction of nearly 75%. This pattern is consistent with substantial improvements in technology and energy efficiency.

The WF region, representing other Asian countries, fluctuated considerably because it aggregates economies at different stages of development. The coefficient peaked at 115,166.79 kg CO₂/million EUR in 2016, then declined to 32,649.92 kg CO₂/million EUR in 2022, above China's level. The aggregate therefore exhibits structural heterogeneity and uneven mitigation.

The EU consequently displayed the highest carbon efficiency, while the United States recorded a particularly large decline in its coefficient. The US coefficient nevertheless remained slightly above China's in 2022. China's initial carbon intensity was comparatively high, but its improvement was rapid. The volatility of other Asian countries illustrates the heterogeneity concealed by regional aggregation, while Japan's trajectory differs from the steadier improvement observed in several other developed economies.

3.3.2 Changes in China's Net Trade-Embodied Carbon

Figure 3.3 shows China's export-embodied carbon, import-embodied carbon, and net trade-embodied carbon, where the net value is the difference between exports and imports.

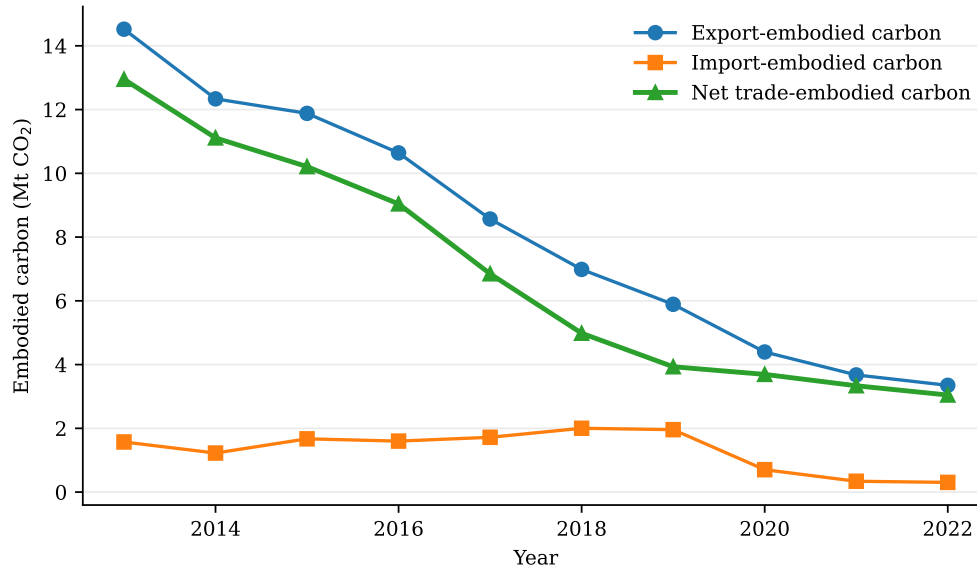


Figure 3.3: Changes in the embodied carbon of China's foreign trade

Both export- and import-embodied emissions declined markedly from 2013 to 2022, but at different rates and by different absolute amounts. Export-embodied carbon fell by 76.9%, from 14.5216 Mt CO₂ in 2013 to 3.3492 Mt CO₂ in 2022. Import-embodied carbon fell by 80.6%, from 1.5723 to 0.3044 Mt CO₂. Despite these substantial reductions, the absolute volume embodied in exports remained far greater than that embodied in imports.

The net value narrowed continuously. It peaked at 12.9494 Mt CO₂ in 2013, declined to approximately 4.9836 Mt CO₂ in 2018, then stabilized at a lower level and reached 3.0448 Mt CO₂ in 2022. China therefore remained a net exporter of embodied carbon throughout the period, but the intensity of its net carbon export weakened considerably.

The trend suggests that the combined effects of lower carbon intensity and export restructuring are gradually reducing China's role as a carbon-intensive producer within global value chains, even though it continues to bear substantial emissions responsibility for internationally consumed goods. The narrowing net balance reflects easing pressure from export-embodied carbon and provides an empirical basis for a more equitable allocation of international carbon responsibility.

3.3.3 Comparison across Trading Partners

Figure 3.4 presents the net trade-embodied carbon between China and the United States, Japan, India, other Asian countries, and the European Union from 2013 to 2022.

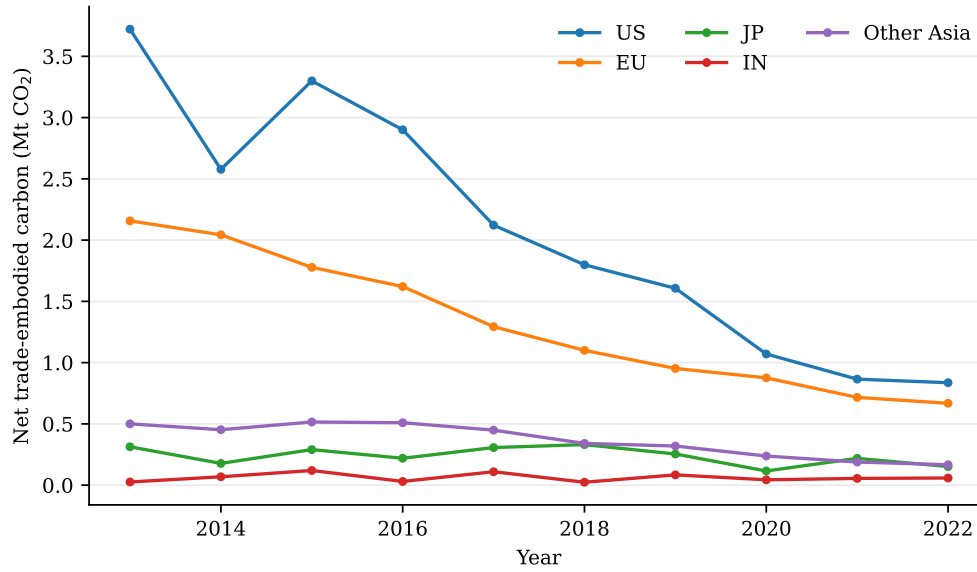


Figure 3.4: Net trade-embodied carbon between China and five trading partners

China's net embodied-carbon exports to the United States and the European Union were consistently the largest and constituted the principal destinations of its transferred emissions. The balance with the United States peaked at 3.7214 Mt CO₂ in 2013 and declined continuously to 0.8361 Mt CO₂ in 2022, a reduction of more than 77%. Nevertheless, carbon embodied in exports accounted for 97.81% of the bilateral total in 2022, indicating that exports to the United States remained a major source of China's trade-related carbon burden.

The balance with the European Union also fell markedly, from 2.1572 Mt CO₂ in 2013 to 0.6681 Mt CO₂ in 2022, a reduction of approximately 69%. Export-embodied carbon accounted for more than 95% of the bilateral total throughout most of the period and 98.68% in 2022, showing that China continued to bear the overwhelming majority of emissions generated by bilateral textile trade.

Other Asian countries ranked third. The balance declined from 0.5003 Mt CO₂ in 2013 to 0.1662 Mt CO₂ in 2022, while the export share rose to 99.31%. Although the total was far below those of the United States and the EU, it was volatile and approached or exceeded 0.5 Mt CO₂ in some years. This instability reflects both the structure of regional carbon flows and the substantial heterogeneity within the aggregate region.

Net values for Japan and India were lower. The balance with Japan declined from 0.3130 Mt CO₂ in 2013 to 0.1519 Mt CO₂ in 2022, while the export-embodied share generally remained above 97%, indicating a comparatively stable bilateral textile-trade structure. The balance with India fluctuated more strongly, increasing from only 0.0258 Mt CO₂ in 2013 to 0.0583 Mt CO₂ in 2022, but remained much smaller than those of the other partners. Its export-embodied share was often below 90% and reached a minimum of 56.75%, indicating that China–India trade was characterized more by two-way complementarity than by a conventional one-directional transfer of carbon responsibility.

China was thus a net exporter of embodied carbon to all five partners throughout the period. This pattern reflects both the export orientation of its textile industry and the international division of production and consumption. The much larger balances with the United States and the EU are consistent with the importance of these consumer markets for Chinese textiles. Balances with India and other Asian countries were smaller and more volatile. The relative importance of imports and exports also differed among relationships: some were dominated by export-embodied carbon, while others involved more substantial two-way flows.

3.4 Chapter Summary

This chapter calculated and compared the import and export structure, carbon-emission coefficients, and trade-embodied carbon of China's textile industry from 2013 to 2022. China remained a net exporter of trade-embodied carbon, although the scale of the net outflow narrowed progressively. The largest balances were associated with textile exports to the United States and the European Union, while the relationships with other Asian countries, Japan, and India involved smaller or more variable transfers. The industry's carbon-emission coefficient declined markedly, but the reduction in total embodied emissions was more limited.

Chapter 4

Decomposition of Changes in Net Trade-Embodied Carbon

The preceding chapter quantified trade-embodied carbon in China's textile industry. Its evolution is driven not only by fluctuations in imports and exports but also by interacting economic, structural, and technological factors. This chapter applies the Logarithmic Mean Divisia Index (LMDI) to construct a seven-factor model comprising energy mix, energy efficiency, production technology, import and export structure, and import and export scale. The model decomposes changes in net trade-embodied carbon from 2013 to 2022 and provides a quantitative basis for policy intervention.

4.1 LMDI Decomposition Model

4.1.1 Selection of Factors and Data Sources

Representative studies broadly attribute changes in embodied carbon to technology, structure, and scale, although some extend the framework to capital–energy and labour–energy substitution, final-consumption pull, export-destination structure, and dependence on the digital economy. Most existing decompositions examine total embodied emissions or exports alone and therefore do not capture the two-way nature of carbon flows where imports and exports coexist. A multidimensional model suitable for net carbon transfers is required.

Table 4.1: Representative decomposition frameworks in the literature

Study	Method	Factors	Operational definitions
Yuan [28]	LMDI	technology; structure; scale	sectoral CO ₂ /sectoral output; sectoral exports/manufacturing exports; total manufacturing exports
Cui [11]	LMDI	technology; structure; scale	departmental CO ₂ /departmental output; departmental exports/manufacturing exports; total manufacturing exports
Yang et al. [14]	SDA	carbon intensity; intermediate-input structure; domestic and foreign ports; digital dependence; export structure and scale	sectoral CO ₂ /output; intermediate inputs/output; digital intermediate-inputs/non-digital-sector inputs; sectoral exports/total exports; national exports/base-period exports
Hu and Wu [13]	LMDI	direct-emission efficiency; emissions energy efficiency; technical efficiency and progress; capital–energy and labour–energy substitution; value-added pull	direct CO ₂ /output; CO ₂ /energy input; actual/potential output; updated/original coefficient; capital/energy; labour/energy; and final-use-induced value added/aggregate value added
Han and Rong [12]	LMDI	trade scale; trade share; industrial structure; carbon technology	trade share; bilateral manufacturing trade; national/bilateral trade; sectoral CO ₂ /output
Zhao et al. [16]	SDA	energy; production technology; trade structure; final demand	emissions per unit of energy; energy mix; energy intensity; intermediate inputs/output; intermediate and final-goods export shares; consumption and capital formation

Study	Method	Factors	Operational definitions
Pu et al. [15]	SDA	technology; structure; scale	departmental CO ₂ /output; departmental/total exports; total exports
Wang et al. [19]	SDA	technology; production structure; final demand; export demand	departmental CO ₂ /output; intermediate inputs/output; domestic final demand; total final demand; exports/departmental demand

Considering the characteristics of China's textile industry and its foreign trade, seven factors measurable with EXIOBASE are selected (Table 4.2).

Table 4.2: Factors used in the LMDI-based index analysis

Factor	Symbol	Definition	Data field
Energy-mix effect	es	Energy-related CO ₂ as a share of total CO ₂	energy share
Energy-efficiency effect	ee	Energy-related CO ₂ per unit of export value	energy emissions / exports
Technology effect	$tech$	CO ₂ per unit of total output	total-emission coefficient
Export-structure effect	xs	Exports as a share of total output	exports / total output
Export-scale effect	xq	Export value	exports
Import-structure effect	ms	Imports as a share of total output	imports / total output
Import-scale effect	mq	Import value	imports

The energy-mix effect captures changes in the share of energy-related emissions and therefore changes in the relative use of cleaner and carbon-intensive energy. The energy-efficiency effect measures energy-related emissions per unit of exports. The technology effect measures carbon efficiency over the entire production process and is closely associated with low-carbon processes and equipment renewal. Export and import structure represent the respective trade values as shares of total output, while export and import scale measure their absolute values. Together, the seven factors form an integrated energy–production–trade system.

4.1.2 LMDI-Based Index-Decomposition Formulation

For any factor z , its contribution between base year 0 and terminal year T is:

$$\Delta F_z = L(F_{\text{net}}^T, F_{\text{net}}^0) \ln \left(\frac{z^T}{z^0} \right), \quad L(a, b) = \frac{a - b}{\ln a - \ln b}, \quad (4.1)$$

where L is the logarithmic mean, carrying the unit of net trade-embodied carbon, and the logarithmic ratio is dimensionless. The weighting term maps the relative change in each efficiency or share variable into an emissions contribution.

The observed change is:

$$\Delta F_{\text{net}} = F_{\text{net}}^T - F_{\text{net}}^0, \quad (4.2)$$

and is decomposed as:

$$\Delta F_{\text{net}} = \Delta F_{es} + \Delta F_{ee} + \Delta F_{tech} + \Delta F_{xs} + \Delta F_{xq} + \Delta F_{ms} + \Delta F_{mq} + R, \quad (4.3)$$

Here R is the reconciliation residual. It is required because net trade-embodied carbon is the difference between export- and import-embodied emissions rather than a single multiplicative identity of the seven selected factors. Accordingly, the reported components are diagnostic index contributions, not a perfect additive or causal decomposition. Each component is calculated from Equation (4.1) using the corresponding factor es , ee , $tech$, xs , xq , ms , or mq . An increase in the share of low-carbon energy is expected to produce a negative energy-mix contribution; improved energy efficiency or production technology should likewise reduce the balance. Changes in trade structure and absolute trade values determine the structure and scale effects. Import-related effects must be interpreted with particular care because imports are subtracted when the net balance is calculated.

4.2 Results and Discussion

4.2.1 Contributions to Changes in Net Trade-Embodied Carbon

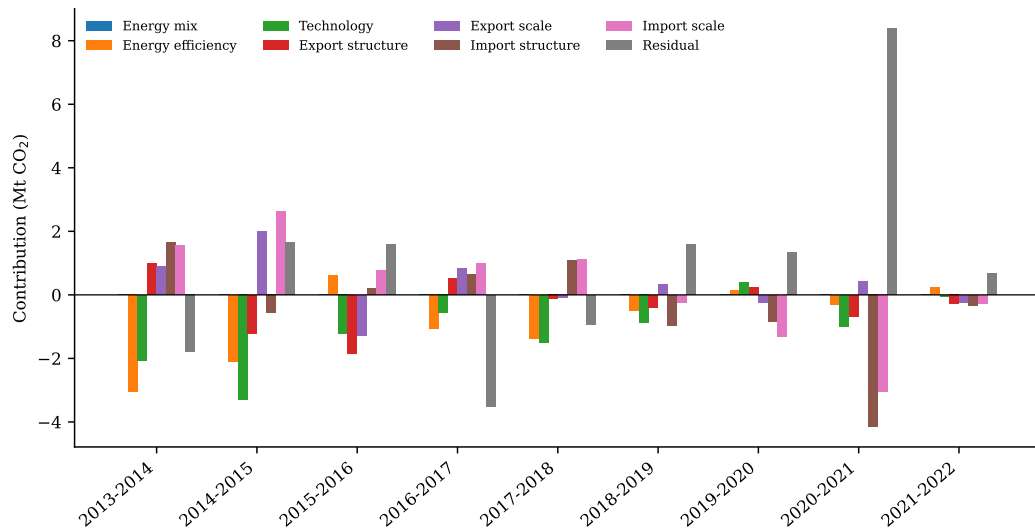


Figure 4.1: Contributions to changes in net trade-embodied carbon

Figure 4.1 shows an overall decline in China's net trade-embodied carbon from 2013 to 2022, with substantial annual variation in both magnitude and estimated drivers. Technology and energy efficiency were frequently negative, whereas structure and scale effects changed sign across periods. Because the specification includes a reconciliation residual, the components should be used to compare direction and relative prominence rather than treated as an exact partition of the observed annual change.

The residual is material in several periods. It was -1.80 Mt CO₂ in 2013–2014 and -3.52 Mt CO₂ in 2016–2017. It reached $+8.38$ Mt CO₂ in 2020–2021. These values confirm that the seven-factor specification does not provide a perfect additive decomposition of

the net balance. The factor estimates are therefore exploratory and should not be interpreted as causal effects.

The energy-mix effect remained small throughout. First, adjustment of the textile industry's energy mix was limited: fossil fuels, particularly coal, remained dominant, while the share of clean energy was comparatively low. The emissions factor per unit of energy therefore changed only modestly. Second, textiles are a light-industry category with lower energy intensity and a relatively stable energy mix, so this factor had only a weak marginal effect on transfers of carbon responsibility.

Net embodied carbon declined by 1.84 Mt CO₂ in 2013–2014 and 0.90 Mt CO₂ in 2014–2015. Trade expansion was pronounced: the export-scale contributions were +0.91 and +2.00 Mt CO₂, while the import-scale contributions reached +1.56 and +2.64 Mt CO₂, respectively. Thus, even as carbon intensity declined, continued trade expansion exerted strong upward pressure on embodied emissions.

The largest decline occurred in 2016–2017, when the balance fell by 2.19 Mt CO₂. Energy efficiency and technology made negative index contributions of –1.08 and –0.57 Mt CO₂, respectively. The trade-related components were mixed: export structure, export scale, import structure, and import scale were all positive in the reported index calculations.

In 2019–2020 and 2020–2021, import- and export-related effects moved in different directions. In 2020–2021, the export-scale contribution was +0.43 Mt CO₂, whereas the import-scale contribution was –3.06 Mt CO₂. Technology and energy efficiency were also negative in that period. These results indicate that the export recovery created upward pressure while import-side changes reduced the net balance.

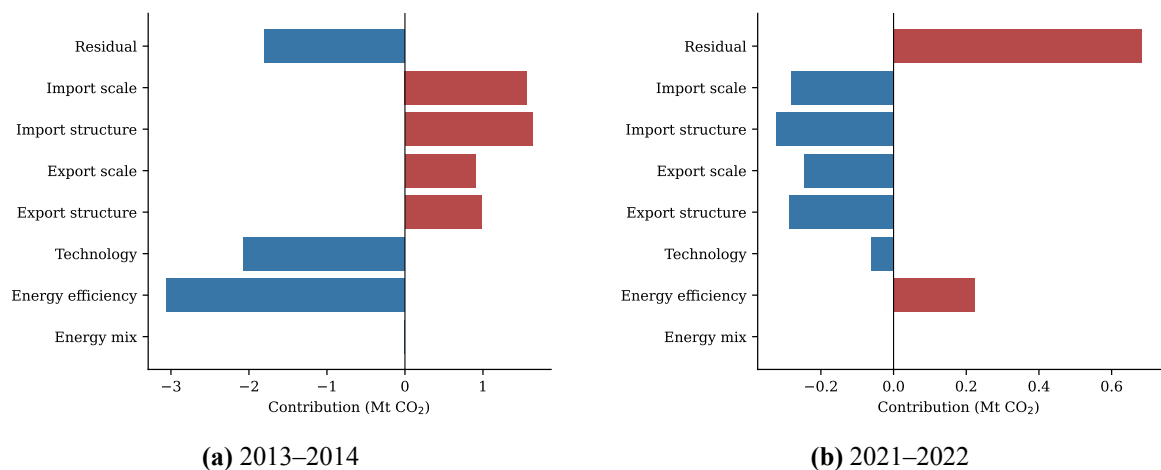


Figure 4.2: Factor contributions in selected periods

Although net trade-embodied carbon declined in both selected periods, the estimated mechanisms differed. In 2013–2014, the balance fell by 1.84 Mt CO₂. Energy efficiency and technology contributed –3.06 and –2.07 Mt CO₂, respectively, while export structure, export scale, import structure, and import scale jointly produced positive index contributions

of approximately 5.09 Mt CO₂. The direction of the estimates therefore suggests efficiency gains offsetting pressure from trade structure and scale, subject to the reconciliation residual described above.

In 2021–2022, the balance fell by only 0.29 Mt CO₂. Energy efficiency was the sole positive index contribution (+0.23), while the other six factors were negative. Import structure and scale made the largest negative contributions, at –0.32 and –0.28 Mt CO₂, followed by export structure and scale at –0.29 and –0.25 Mt CO₂. Technology and energy-mix contributions were small.

The drivers thus changed over time. Early reductions relied primarily on rapid improvements in energy efficiency and technology, whereas later reductions were driven more by import restructuring, import contraction, and moderation of export activity.

4.2.2 Partner-Specific Contributions

Table 4.3: LMDI-based index contributions for trade with the United States (Mt CO₂)

Period	ΔF	es	ee	$tech$	xs	xq	ms	mq
2013–2014	-1.1439	-0.0012	-0.5117	-0.5374	-0.0269	-0.0460	-0.5899	-0.6090
2014–2015	0.7221	-0.0001	-1.0092	-0.9090	0.1002	0.9824	0.2782	1.1605
2015–2016	-0.3992	-0.0010	0.1904	-0.3970	-0.5884	-0.4046	-0.0042	0.1796
2016–2017	-0.7784	-0.0003	-0.0015	-0.1805	-0.1793	-0.0734	2.2202	2.3261
2017–2018	-0.3235	-0.0014	-0.6463	-0.5023	0.1427	0.1532	0.0789	0.0895
2018–2019	-0.1917	-0.0012	-0.3893	-0.3400	0.0481	0.3305	-0.9431	-0.6607
2019–2020	-0.5367	-0.0008	0.2361	0.1376	-0.0993	-0.2637	-0.4883	-0.6527
2020–2021	-0.2050	0.0023	-0.0537	-0.2704	-0.2144	0.0888	-0.7571	-0.4539
2021–2022	-0.0290	-0.0000	0.0180	-0.0165	-0.0346	-0.0235	-0.1074	-0.0963

Note: The table preserves the seven reported index contributions. The reconciliation residual $R = \Delta F - \sum_z \Delta F_z$ is not shown in the table and must be included when assessing additivity. The cumulative net balance with the United States declined by 2.89 Mt CO₂. Energy efficiency (−2.17) and technology (−3.02) were the principal mitigating factors, demonstrating sustained optimization of the Chinese production chain serving the US market. Trade expansion, led by the import-scale contribution (+1.28), offset part of these gains; persistent US demand for Chinese products therefore remained an important source of carbon pressure.

The balance with the EU declined cumulatively by 1.49 Mt CO₂. Energy efficiency (−1.65) and technology (−1.93) contributed substantially, while export structure (−0.29) and import structure (−1.05) were also favourable. The export-scale contribution increased by +0.76, but did not reverse the overall decline.

The balance with Japan declined by only 0.16 Mt CO₂, the smallest change. Energy efficiency (−0.38) and technology (−0.33) were the principal negative effects, while export scale (+0.22) and structure (+0.05) were modestly positive. Import-related effects were small, reflecting a stable bilateral textile-trade structure.

India was the only partner for which China’s net embodied-carbon export increased, by 0.03 Mt CO₂. Energy efficiency (−0.12) and technology (−0.08) reduced the balance, but export scale (+0.09) and export structure (+0.04) increased it, indicating that export growth generated a modest carbon rebound.

The balance with other Asian countries declined by 0.33 Mt CO₂. Energy efficiency (−0.57) and technology (−0.53) were the principal negative index contributions, while export scale remained positive (+0.32). Import structure (−0.32) and import scale (−0.04) were also negative, producing a volatile pattern in which technical improvement and trade expansion interacted.

Across all partners, export scale frequently increased net embodied carbon, reflecting the industry’s dependence on export-volume growth. Technology and energy efficiency generally reduced the balance, most strongly in trade with the United States and EU, but did not always offset expansion. Structural effects differed across partners and periods. The estimated direction was less stable for India and other Asian countries than for the United States and EU. Energy-mix effects were small in most relationships but comparatively more volatile for India and other Asian countries.

4.3 Policy Recommendations

- (1) Because export scale is the principal driver of increased embodied carbon, China should preserve the international competitiveness of its textile industry while moderating growth in exports of carbon-intensive products, optimizing the composition of aggregate exports, and shifting from volume-led to quality-led growth.

- (2) Technology and energy efficiency offset carbon growth in many periods, but their aggregate effect remains limited. Policy and fiscal support should be strengthened for green production processes, energy-efficient equipment, and intelligent manufacturing.
- (3) Partner-specific mechanisms require differentiated approaches. For technology-led markets such as the United States and EU, China should strengthen low-carbon standards and certification and participate actively in the formulation of green-trade rules. For India and other Asian markets, cooperation on low-carbon production, technology, and efficiency should be strengthened.
- (4) Given the role of imports in reducing net embodied carbon, transparency of carbon information at the border should be improved, and lower-carbon raw materials and intermediate goods should replace carbon-intensive inputs where feasible.

More specifically, the carbon intensity of textile exports to the United States should be reduced through product upgrading and cleaner production. For the EU, where carbon requirements for imports are stringent, priority should be given to technical upgrading, green-standard compliance, and regulatory alignment. The stable, low-balance relationship with Japan should be maintained while expanding cooperation in energy-saving technology. For volatile trade with India, exports should shift towards higher-value, lower-carbon products, and carbon-intensive imports should be subject to appropriate monitoring. Cooperation with other Asian partners should accelerate the diffusion of low-carbon technology and cleaner production.

4.4 Chapter Summary

Using an LMDI-based index framework, this chapter assessed changes in net trade-embodied carbon across seven dimensions: energy mix, energy efficiency, production technology, import and export structure, and import and export scale. Export scale frequently increased the balance, whereas technology and energy efficiency generally reduced it. The estimated structure of drivers differed among partners. Because the net balance is not a single multiplicative identity of the seven factors, the results include a reconciliation residual and should be interpreted as diagnostic contributions.

Chapter 5

Conclusions and Outlook

5.1 Conclusions

Using environmentally extended input-output data from EXIOBASE 3 for 2013–2022, this study accounted for emissions in China’s textile industry and developed coefficient-based estimates of the carbon embodied in its foreign trade. An LMDI-based index framework was then used to examine the principal determinants associated with changes in the net balance. The main conclusions are as follows.

- (1) The carbon-emission coefficient of China’s textile industry declined markedly from 71,343 to 18,747 kg CO₂/million EUR, indicating an overall improvement in carbon efficiency and the initial effectiveness of policies promoting green manufacturing, energy restructuring, technological innovation, and energy efficiency. The largest reduction occurred from 2013 to 2018, when the coefficient reached 27,843 kg CO₂/million EUR, principally because of stronger policy intervention and technology-renewal cycles. China remained a net exporter of embodied carbon throughout the study period. Export-embodied carbon accounted for 91.67% of the bilateral total in 2022; even its minimum share, 75.05% in 2019, remained high. Carbon embodied in exported textiles and apparel therefore far exceeded that embodied in imports, demonstrating the substantial emissions responsibility borne by China in global supply chains.
- (2) Significant differences existed among the five trading partners. The United States and EU were the principal markets for Chinese textiles and produced the largest net embodied-carbon balances. In 2022, the balances were 0.8361 Mt CO₂ for the United States and 0.6681 Mt CO₂ for the EU. Net transfers to India and other Asian countries were smaller. The balance with India was 0.0583 Mt CO₂ in 2022, when exports accounted for 88.80% of total bilateral embodied carbon, compared with only 56.75% in 2013.

- (3) Seven index effects were estimated: energy mix, energy efficiency, technology, export structure, export scale, import structure, and import scale. Export scale made positive contributions of +0.9070 Mt CO₂ in 2013–2014 and +1.9968 Mt CO₂ in 2014–2015. Energy efficiency contributed –3.0570 Mt CO₂ in 2013–2014, while technology contributed –3.3123 Mt CO₂ in 2014–2015. Import and export effects varied dynamically across periods. The import-scale effect shifted from +2.6423 Mt CO₂ in 2014–2015 to –3.0558 Mt CO₂ in 2020–2021, while the import-structure contribution was –4.1601 Mt CO₂ in 2020–2021. These values are diagnostic index contributions and must be interpreted together with the reconciliation residual.

5.2 Limitations and Future Research

Although the study provides a relatively comprehensive accounting and analysis, two principal limitations remain.

- (1) The coefficient-based trade calculation does not fully trace upstream production linkages across the complete multi-regional supply chain. The estimates may therefore omit part of the indirect carbon transferred through intermediate inputs. Future work should use the full Leontief structure of an EE-MRIO model to capture these flows more accurately.
- (2) Although the decomposition covers scale, structure, and technology, it does not adequately account for development of the digital economy, the intensity of policy intervention, or changes in consumer behaviour. Future research should expand the set of determinants and combine qualitative and quantitative evidence to improve analytical completeness and the operational relevance of policy recommendations.
- (3) Net trade-embodied carbon is the difference between export- and import-embodied emissions and is not a single multiplicative identity of the seven selected factors. The present index contributions therefore require a reconciliation residual. Future studies should decompose export and import footprints separately and combine them only after obtaining internally consistent additive results.

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Appendix A

Data Tables

Table A.1: Bilateral textile trade between China and five regions (million EUR)

Year	Region	Exports	Imports	Trade balance
2013	US	35698.6211	3741.8993	31956.7218
2013	JP	3119.3419	2790.8614	328.4805
2013	IN	942.7332	1025.5058	-82.7726
2013	WF	4184.7239	3565.8113	618.9126
2013	EU	22558.1635	6499.6830	16058.4805
2014	US	35175.3047	2699.9217	32475.3829
2014	JP	3785.1302	3344.2695	440.8607
2014	IN	1284.7202	1563.7022	-278.9819
2014	WF	5841.6713	4021.5510	1820.1203
2014	EU	26172.5037	6636.2410	19536.2627
2015	US	49222.9099	3430.8214	45792.0885
2015	JP	4104.3581	3650.1609	454.1972
2015	IN	2582.5922	1861.9640	720.6281
2015	WF	7220.7029	4735.9766	2484.7264
2015	EU	29124.2446	5991.9180	23132.3266
2016	US	43192.5880	4138.1392	39054.4488
2016	JP	3298.9044	3419.1107	-120.2063
2016	IN	1774.9922	1798.2097	-23.2175
2016	WF	6802.3251	3999.6302	2802.6949
2016	EU	25850.8750	6011.6235	19839.2514
2017	US	41938.7104	3659.8373	38278.8731
2017	JP	6161.8079	3519.5943	2642.2137
2017	IN	2604.7980	3517.4728	-912.6747
2017	WF	8721.1427	5092.4212	3628.7215

Year	Region	Exports	Imports	Trade balance
2017	EU	29303.7068	8188.9130	21114.7938
2018	US	45355.6342	2101.0950	43254.5392
2018	JP	9211.9770	2752.6853	6459.2917
2018	IN	1888.9117	2234.2182	-345.3065
2018	WF	7710.1915	6259.9942	1450.1973
2018	EU	29909.2541	7034.9660	22874.2881
2019	US	55082.9942	350.9328	54732.0614
2019	JP	7320.0567	4325.8577	2994.1990
2019	IN	3378.3410	1986.5213	1391.8198
2019	WF	8162.4742	5970.3172	2192.1570

Table A.2: Bilateral textile trade between China and five regions (million EUR)

Year	Region	Exports	Imports	Trade balance
2019	EU	32867.2068	7885.3489	24981.8579
2020	US	45111.7561	308.7220	44803.0342
2020	JP	3914.3801	5970.6259	-2056.2458
2020	IN	2192.7840	4058.5564	-1865.7724
2020	WF	8500.8002	6910.9804	1589.8198
2020	EU	39940.5415	8233.7554	31706.7861
2021	US	49466.7142	4928.9725	44537.7416
2021	JP	10903.3690	16465.1425	-5561.7736
2021	IN	3713.1614	4092.0488	-378.8874
2021	WF	8949.5288	5921.2145	3028.3143
2021	EU	39844.8422	16360.8328	23484.0094
2022	US	48118.9752	6172.7018	41946.2735
2022	JP	7553.3493	15607.0505	-8053.7012
2022	IN	4008.1121	3198.5268	809.5853
2022	WF	7990.5316	4664.5007	3326.0309
2022	EU	38027.6514	15577.0814	22450.5700

Table A.3: Total embodied carbon of China's textile industry (Mt CO₂)

Year	Embodied carbon
2013	67.9953
2014	56.8687

Year	Embodied carbon
2015	56.3519
2016	52.6022
2017	51.0513
2018	39.7030
2019	38.3803
2020	37.6106
2021	38.9135
2022	38.6659

Table A.4: Embodied carbon in China's foreign trade (Mt CO₂)

Year	Export embodied	Import embodied	Net embodied	Export share
2013	14.5216	1.5722	12.9493	90.23
2014	12.3361	1.2246	11.1115	90.97
2015	11.8807	1.6706	10.2101	87.67
2016	10.6406	1.5995	9.0410	86.93
2017	8.5667	1.7185	6.8482	83.29
2018	6.98540	2.0018	4.9835	77.73

Table A.5: Embodied carbon in China's foreign trade (Mt CO₂)

Year	Export embodied	Import embodied	Net embodied	Export share
2019	5.89069	1.9586	3.9320	75.05
2020	4.39525	0.7030	3.6921	86.21
2021	3.67549	0.3396	3.3358	91.54
2022	3.34917	0.3044	3.0447	91.67

Table A.6: Trade-embodied carbon between China and five regions (Mt CO₂)

Year	Region	Export embodied	Import embodied	Net embodied
2013	US	3.7665	0.0451	3.7214
2013	JP	0.3195	0.0065	0.3130
2013	IN	0.0484	0.0226	0.0258
2013	WF	0.5071	0.0067	0.5003
2013	EU	2.1885	0.0313	2.1572

Year	Region	Export embodied	Import embodied	Net embodied
2014	US	2.5879	0.0103	2.5775
2014	JP	0.1847	0.0073	0.1775
2014	IN	0.0822	0.0140	0.0682
2014	WF	0.4571	0.0048	0.4523
2014	EU	2.0800	0.0368	2.0433
2015	US	3.3118	0.0122	3.2996
2015	JP	0.2960	0.0059	0.2901
2015	IN	0.1298	0.0101	0.1197
2015	WF	0.5274	0.0120	0.5154
2015	EU	1.8193	0.0414	1.7779
2016	US	2.9109	0.0105	2.9005
2016	JP	0.2269	0.0076	0.2193
2016	IN	0.0414	0.0113	0.0301
2016	WF	0.5356	0.0262	0.5094
2016	EU	1.6672	0.0464	1.6208
2017	US	2.1373	0.0152	2.1221
2017	JP	0.3135	0.0068	0.3068
2017	IN	0.1199	0.0104	0.1095
2017	WF	0.4747	0.0257	0.4490
2017	EU	1.3514	0.0580	1.2934
2018	US	1.8204	0.0218	1.7986
2018	JP	0.3414	0.0104	0.3310
2018	IN	0.0428	0.0191	0.0237
2018	WF	0.3525	0.0118	0.3407
2018	EU	1.1752	0.0751	1.1002
2019	US	1.6252	0.0183	1.6069

Table A.7: Trade-embodied carbon between China and five regions (Mt CO₂)

Year	Region	Export embodied	Import embodied	Net embodied
2019	JP	0.2635	0.0091	0.2543
2019	IN	0.0937	0.0097	0.0840
2019	WF	0.3277	0.0085	0.3193
2019	EU	1.0191	0.0665	0.9526
2020	US	1.0745	0.0043	1.0702
2020	JP	0.1242	0.0086	0.1156
2020	IN	0.0485	0.0050	0.0435

Year	Region	Export embodied	Import embodied	Net embodied
2020	WF	0.2438	0.0061	0.2377
2020	EU	0.9439	0.0681	0.8757
2021	US	0.8804	0.0153	0.8652
2021	JP	0.2210	0.0023	0.2187
2021	IN	0.0595	0.0044	0.0551
2021	WF	0.1911	0.0023	0.1888
2021	EU	0.7279	0.0116	0.7164
2022	US	0.8549	0.0188	0.8361
2022	JP	0.1542	0.0022	0.1519
2022	IN	0.0627	0.0043	0.0583
2022	WF	0.1673	0.0011	0.1662
2022	EU	0.6772	0.0091	0.6681

Table A.8: Definitions of EXIOBASE-derived variables

Variable	Definition
F_export	Export-embodied carbon (kg)
F_import	Import-embodied carbon (kg)
F_net	Net trade-embodied carbon
E_total	Total energy-related carbon emissions (kg)
E_energy	Emissions attributable to energy consumption (kg)
X_export	Total exports (million EUR)
X_import	Total imports (million EUR)
X_total	Total industry output (million EUR)
e_total	Carbon-emission coefficient (kg/million EUR) = CO2 emissions/total output
energy_share	Energy-related emissions/total emissions

Note on decomposition tables: The following tables preserve the seven index contributions reported in the original analysis. Because net trade-embodied carbon is not a single multiplicative identity of these factors, the entries do not sum exactly to ΔF . The reconciliation residual is $R = \Delta F - \sum_z \Delta F_z$ and should be considered when interpreting the results.

Table A.9: LMDI-based index contributions to changes in net trade-embodied carbon (Mt CO₂)

Period	ΔF	<i>es</i>	<i>ee</i>	<i>tech</i>	<i>xs</i>	<i>xq</i>	<i>ms</i>	<i>mq</i>
2013-2014	-1.8378	-0.0044	-3.0570	-2.0719	0.9807	0.9070	1.6384	1.5647
2014-2015	-0.9014	-0.0005	-2.0946	-3.3123	-1.2182	1.9968	-0.5727	2.6423
2015-2016	-1.1691	-0.0030	0.6131	-1.2329	-1.8490	-1.2781	0.2127	0.7836
2016-2017	-2.1928	-0.0009	-1.0778	-0.5719	0.5050	0.8406	0.6498	0.9855
2017-2018	-1.8647	-0.0041	-1.3822	-1.5065	-0.1284	-0.0968	1.0746	1.1062
2018-2019	-1.0515	-0.0032	-0.4826	-0.8869	-0.4075	0.3291	-0.9650	-0.2285
2019-2020	-0.2399	-0.0023	0.1528	0.3972	0.2421	-0.2323	-0.8361	-1.3105
2020-2021	-0.3563	0.0084	-0.3067	-0.9847	-0.6696	0.4347	-4.1601	-3.0558
2021-2022	-0.2911	-0.0002	0.2251	-0.0619	-0.2872	-0.2456	-0.3235	-0.2820

Table A.10: LMDI-based index contributions for the European Union (Mt CO₂)

Period	ΔF	<i>es</i>	<i>ee</i>	<i>tech</i>	<i>xs</i>	<i>xq</i>	<i>ms</i>	<i>mq</i>
2013–2014	-0.1139	-0.0008	-0.6880	-0.3623	0.3249	0.3120	0.5323	0.5194
2014–2015	-0.2654	-0.0001	-0.2213	-0.5930	-0.3718	0.2038	-0.2024	0.3732
2015–2016	-0.1571	-0.0005	0.0850	-0.2178	-0.3033	-0.2025	0.0758	0.1767
2016–2017	-0.3274	-0.0002	-0.2255	-0.1051	0.1202	0.1819	0.1880	0.2497
2017–2018	-0.1932	-0.0008	-0.3255	-0.3067	0.0180	0.0244	0.4440	0.4505
2018–2019	-0.1476	-0.0007	-0.1321	-0.2048	-0.0735	0.0966	-0.2026	-0.0325
2019–2020	-0.0769	-0.0006	-0.1972	0.0952	0.2918	0.1781	-0.0305	-0.1442
2020–2021	-0.1594	0.0019	0.0308	-0.2225	-0.2515	-0.0019	-1.6312	-1.3816
2021–2022	-0.0483	-0.0000	0.0278	-0.0134	-0.0413	-0.0323	-0.2258	-0.2167

Table A.11: LMDI-based index contributions for Japan (Mt CO₂)

Period	ΔF	es	ee	$tech$	xs	xq	ms	mq
2013–2014	-0.1355	-0.0001	-0.0890	-0.0412	0.0477	0.0462	0.0192	0.0178
2014–2015	0.1127	-0.0000	-0.0207	-0.0713	-0.0506	0.0186	-0.0314	0.0378
2015–2016	-0.0708	-0.0001	0.0378	-0.0325	-0.0703	-0.0553	0.0530	0.0681
2016–2017	0.0874	-0.0000	-0.1706	-0.0189	0.1517	0.1628	-0.0085	0.0025
2017–2018	0.0242	-0.0002	-0.2085	-0.0818	0.1264	0.1282	0.0540	0.0557
2018–2019	-0.0766	-0.0002	0.0568	-0.0582	-0.1152	-0.0669	-0.0526	-0.0043
2019–2020	-0.1387	-0.0001	0.1065	0.0183	-0.0883	-0.1102	0.0122	-0.0097
2020–2021	0.1031	0.0004	-0.1598	-0.0454	0.1148	0.1657	-0.1866	-0.1357
2021–2022	-0.0668	-0.0000	0.0661	-0.0036	-0.0697	-0.0673	-0.0119	-0.0095

Table A.12: LMDI-based index contributions for India (Mt CO₂)

Period	ΔF	<i>es</i>	<i>ee</i>	<i>tech</i>	<i>xs</i>	<i>xq</i>	<i>ms</i>	<i>mq</i>
2013–2014	0.0424	-0.0000	-0.0213	-0.0075	0.0138	0.0135	-0.0270	-0.0273
2014–2015	0.0515	-0.0000	-0.0648	-0.0285	0.0363	0.0639	-0.0007	0.0269
2015–2016	-0.0896	-0.0000	0.0199	-0.0083	-0.0282	-0.0243	0.0016	0.0054
2016–2017	0.0793	-0.0000	-0.0254	-0.0045	0.0210	0.0236	-0.0010	0.0016
2017–2018	-0.0857	-0.0000	0.0039	-0.0144	-0.0183	-0.0180	0.0206	0.0209
2018–2019	0.0603	-0.0000	-0.0294	-0.0095	0.0198	0.0277	-0.0196	-0.0117
2019–2020	-0.0405	-0.0000	0.0253	0.0064	-0.0189	-0.0266	-0.0346	-0.0423
2020–2021	0.0116	0.0001	-0.0241	-0.0138	0.0104	0.0258	-0.0232	-0.0077
2021–2022	0.0032	-0.0000	-0.0047	-0.0011	0.0036	0.0043	-0.0034	-0.0027

Table A.13: LMDI-based index contributions for other Asian countries (Mt CO₂)

Period	ΔF	es	ee	$tech$	xs	xq	ms	mq
2013–2014	-0.0481	-0.0002	-0.2440	-0.0821	0.1617	0.1588	0.0907	0.0878
2014–2015	0.0632	-0.0000	-0.1068	-0.1502	-0.0434	0.1024	-0.0989	0.0469
2015–2016	-0.0061	-0.0002	-0.0049	-0.0657	-0.0610	-0.0306	-0.0024	0.0281
2016–2017	-0.0604	-0.0001	-0.1333	-0.0347	0.0986	0.1189	0.2555	0.2759
2017–2018	-0.1083	-0.0003	-0.0506	-0.1007	-0.0505	-0.0483	-0.0588	-0.0566
2018–2019	-0.0214	-0.0002	-0.0302	-0.0659	-0.0360	0.0188	-0.0450	0.0098
2019–2020	-0.0816	-0.0002	-0.0170	0.0288	0.0456	0.0112	-0.0415	-0.0759
2020–2021	-0.0490	0.0005	-0.0032	-0.0595	-0.0559	0.0109	-0.2849	-0.2181
2021–2022	-0.0226	-0.0000	0.0189	-0.0034	-0.0224	-0.0201	-0.1351	-0.1328

Appendix B

Data-Processing Code

The following cleaned script documents the core data-processing workflow used for the study. It corrects line wrapping and indentation lost when the original Word appendix was extracted. The EXIOBASE data directory must be configured before execution. The original transcription is retained separately as `analysis_code.py` for comparison.

```
"""Core data-processing workflow for the thesis.

Set DATA_DIR to the directory containing the annual EXIOBASE archives
before
running this script. Monetary values retain the units used by EXIOBASE.
"""

from pathlib import Path

import numpy as np
import pandas as pd
import pymrio

DATA_DIR = Path("/path/to/exiobase_data")
YEARS = range(2013, 2023)

TEXTILE_SECTORS = [
    "Cultivation of plant-based fibers",
    "Wool, silk-worm cocoons",
    "Manufacture of textiles (17)",
    "Manufacture of wearing apparel; dressing and dyeing of fur (18)",
    (
        "Tanning and dressing of leather; manufacture of luggage, handbags,
        "
        "saddlery, harness and footwear (19)"
    ),
    "Incineration of waste: Textiles",
```

```

    "Landfill of waste: Textiles",
]

CO2_FLOWS = [
    "CO2 - combustion - air",
    "CO2_bio - combustion - air",
    "CO2 - waste - fossil - air",
    "CO2 - waste - biogenic - air",
]

EU_COUNTRIES = {
    "AT", "BE", "BG", "HR", "CY", "CZ", "DK", "EE", "FI", "FR", "DE",
    "GR", "HU", "IE", "IT", "LV", "LT", "LU", "MT", "NL", "PL", "PT",
    "RO", "SK", "SI", "ES", "SE",
}

PARTNER_REGIONS = {
    "United States": {"US"},
    "Japan": {"JP"},
    "India": {"IN"},
    "European Union": EU_COUNTRIES,
    "Other Asian Countries": {"WF"},
}

def load_year(year):
    archive = DATA_DIR / f"IOT_{year}_ixi.zip"
    mrio = pymrio.parse_exiobase3(archive)
    mrio.calc_all()
    return mrio

def selected_emissions(mrio):
    """Return direct CO2 emissions by region and sector in kilograms."""
    emissions = None
    for flow in CO2_FLOWS:
        values = mrio.air_emissions.get_row_data(flow)["F"].iloc[:, 0]
        emissions = values if emissions is None else emissions.add(
            values, fill_value=0
        )
    return emissions

def textile_mask(index, regions=None):
    mask = index.get_level_values("sector").isin(TEXTILE_SECTORS)
    if regions is not None:
        mask &= index.get_level_values("region").isin(regions)

```

```

    return mask

def direct_carbon_coefficients(mrio):
    """Calculate direct coefficients in kg CO2 per monetary output unit."""
    emissions = selected_emissions(mrio)
    output = mrio.x.squeeze()
    coefficients = emissions.div(output).replace([np.inf, -np.inf], np.nan)
    return coefficients.fillna(0)

def final_demand_trade(mrio, producer_regions, consumer_regions):
    """Return textile final demand supplied by one region group to another
    ."""
    row_mask = textile_mask(mrio.Y.index, producer_regions)
    columns = [
        column for column in mrio.Y.columns
        if column[0] in consumer_regions
    ]
    return mrio.Y.loc[row_mask, columns].sum(axis=1)

def embodied_trade_by_partner(mrio):
    coefficients = direct_carbon_coefficients(mrio)
    results = []

    for partner, regions in PARTNER_REGIONS.items():
        exports = final_demand_trade(mrio, {"CN"}, regions)
        imports = final_demand_trade(mrio, regions, {"CN"})

        export_carbon = coefficients.reindex(exports.index).mul(exports).
            sum()
        import_carbon = coefficients.reindex(imports.index).mul(imports).
            sum()

        results.append(
            {
                "partner": partner,
                "export_value": exports.sum(),
                "import_value": imports.sum(),
                "export_carbon_kg": export_carbon,
                "import_carbon_kg": import_carbon,
                "net_carbon_kg": export_carbon - import_carbon,
            }
        )

```



```

return pd.DataFrame(results).set_index("partner")

def production_based_textile_emissions(mrio):
    """Sum production-based textile emissions for China."""
    rows = mrio.air_emissions.D_pba.index.isin(CO2_FLOWS)
    columns = textile_mask(mrio.air_emissions.D_pba.columns, {"CN"})
    return mrio.air_emissions.D_pba.loc[rows, columns].to_numpy().sum()

def logarithmic_mean(current, previous):
    if current <= 0 or previous <= 0:
        return np.nan
    if np.isclose(current, previous):
        return current
    return (current - previous) / (np.log(current) - np.log(previous))

def diagnostic_index_contributions(net_current, net_previous, factors):
    """Calculate factor indices and the residual needed for reconciliation
    .

    ``factors`` maps each factor name to ``(current_value, previous_value)``.
    The net balance is not a single multiplicative identity of these
    factors,
    so the residual must be reported with the individual contributions.
    """
    weight = logarithmic_mean(net_current, net_previous)
    contributions = {
        name: weight * np.log(current / previous)
        for name, (current, previous) in factors.items()
        if current > 0 and previous > 0
    }
    observed_change = net_current - net_previous
    contributions["residual"] = observed_change - sum(contributions.values
    ())
    return contributions

def main():
    annual_trade = []
    annual_production = []

    for year in YEARS:
        mrio = load_year(year)

```

```

trade = embodied_trade_by_partner(mrio).reset_index()
trade.insert(0, "year", year)
annual_trade.append(trade)

annual_production.append(
    {
        "year": year,
        "production_emissions_kg": production_based_textile_emissions
            (
                mrio
            ),
    }
)

pd.concat(annual_trade, ignore_index=True).to_csv(
    "trade_embodied_carbon_by_partner.csv", index=False
)
pd.DataFrame(annual_production).to_csv(
    "china_textile_production_emissions.csv", index=False
)

if __name__ == "__main__":
    main()

```